

TREVOR CROUSE

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EDUCATION

CORNELL UNIVERSITY, Duffield College of Engineering, Ithaca, New York

Bachelor of Science Degree in Mechanical Engineering, GPA 3.0 Expected May 2028

- **Relevant Coursework:** System Dynamics; Statics and Mechanics of Solids; Thermodynamics, Mechanical Design; Mechanics of Engineering Materials; Waves and Quantum Physics; Fluid Mechanics, Engineering Leadership
- **Campus Involvement:** American Society of Mechanical Engineers, Aerodynamic Research, Makers Club, Cornell Daily Sun, Cornell Club Hockey

SPECIALIZED SKILLS AND CERTIFICATIONS

- ANSYS Fluent; CFD; SolidWorks CAD; Fusion 360; NX; MathWorks MATLAB; Python; Aerodynamic Analysis; Mechanical Design; Machine Shop; Microsoft Word/Excel/PowerPoint; Technical Documentation
- AutoCAD Basic Training (Jan. 2026 – Udemy); Airplane Design and Aerodynamic Basic (Oct. 2024 – Udemy); Automobile Fundamentals and Advanced (Oct. 2024 – ValueKnow)

RELEVANT EXPERIENCE

Freudenberg-NOK Sealing Technologies; Bristol, New Hampshire

Mechanical Engineering Intern

May 2026 – August 2026

- Designed fixtures to reduce downtime and improve efficiency, using ANSYS Mechanical FEA to model wear and inform material selection. Performed analysis on tool fixtures to evaluate stress concentrations, predict tool wear, and inform material selection for production runs
- Developed a rust removal process compatible with production hours, implemented on the line and saving over \$12,000 in parts that would otherwise have been scrapped
- Researched and evaluated metal shredder vendors, identifying a solution with a 3.6-year ROI against \$41,000/year in lost clean scrap, putting installation on track for approval next year
- Remodeled die components in NX to support progressive die tooling for stamping operations

Design, Build, Fly Project Team; Cornell University; Ithaca, New York

Aerodynamics and Controls Sub-team Lead

November 2024 – Present

- Directed a 14-member aerodynamic team to create all lifting components of a high-performance RC aircraft. Utilized SolidWorks CAD for design and ANSYS Fluent for aerodynamic performance
- Collaborated with cross-functional sub-team to design and integrate subsystems, using ANSYS Mechanical FEA to perform stress analysis on components and to create simulations of ground competition missions
- Led technical documentation on a 58-page competition report for the annual *AIAA Design, Build, Fly* competition.

PROJECTS

Custom Designed Wind Tunnel; Independent Project

June 2026 – August 2026

- Sized and designed an open-circuit wind tunnel with a 5th-order polynomial contraction, honeycomb flow straightener, and 2×2 fan array, achieving 3.3 m/s airspeed at the center of the test section
- Modeled the full assembly in SolidWorks and 3D printed most components, designing remaining parts around materials already on hand to minimize cost
- Built a test section with acrylic viewing windows and a laminar fog generator to enable flow visualization around test airfoils

Aircraft Wing Design and Manufacturing for Design, Build, Fly; Ithaca, New York

August 2025 – April 2026

- Optimized airfoil and wing shape around AIAA constraints using XFLR5 and a self-made MATLAB sizing script
- Designed a manufacturable full-scale wing in SolidWorks creating useful assemblies to test aileron deflections and to use in ANSYS CFD to perform lift and drag analysis

Wing Endplate Aerodynamic Research; Ithaca, New York

January 2026 – March 2026

- Conducted research on wing endplates efficiency and contribution to lift at low Reynolds Numbers, using professor insight to design endplate shapes in SolidWorks and using ANSYS CFD to perform lift-drag analysis
- Wrote a report on the effectiveness of wing endplates, balancing their effectiveness while comparing their benefits with weight and manufacturing constraints